

# Software Engineering

## Lab 4 - LLM-Assisted Debugging and Feature Implementation

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Scenario : #14 - 2048 (terminal game)  
Repository : [github.com/RohanMay-Suresh/SE-Labs-PES1UG24CS383/Lab4](https://github.com/RohanMay-Suresh/SE-Labs-PES1UG24CS383/Lab4)

## 1. Deliverables

| Deliverable                              | Link                                                                                                                                                  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated lab repository                   | <a href="https://github.com/RohanMay-Suresh/SE-Labs-PES1UG24CS383/tree/main/Lab4">github.com/RohanMay-Suresh/SE-Labs-PES1UG24CS383/tree/main/Lab4</a> |
| Before video (original code)             | <a href="#">Google Drive - Before video</a>                                                                                                           |
| After video (all changes implemented)    | <a href="#">Google Drive - After video</a>                                                                                                            |
| Prompts used / complete LLM chat history | <a href="https://claude.ai/share/463f54ba-b87c-44e4-aa29-bf588b3ee43b">https://claude.ai/share/463f54ba-b87c-44e4-aa29-bf588b3ee43b</a>               |

The starter code is from [SETAPESU26/14 2048](#). Only `board.py` and `game.py` were changed. No external dependencies or persistent storage were added; all state stays in memory.

| File                      | Changes                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| board.py                  | Merge fix in slide_line(), score and merge counting in the four move functions, has_won(), snapshot() / restore() for undo |
| game.py                   | Unchanged-move handling, one-level undo, best score, feedback messages, exact input validation                             |
| main.py, requirements.txt | Unchanged                                                                                                                  |

Evidence below is real output from running the original starter code and the updated Lab 4 code with the same starting boards (boards were set up deliberately, as the lab README suggests, so each case is shown in one or two moves).

## 2. Task 1 - Correct Merge Semantics (bug fix)

**Bug reproduced:** in the original code a tile created by a merge could merge again in the same move. The row 2 2 4 8 chain-merged into a single 16 ( $2+2=4$ ,  $4+4=8$ ,  $8+8=16$ ). The score also stayed at 0, because merges never added points.

**Original code - start**

Score: 0    Best: 0

**Original code - > a (left)**

Score: 0    Best: 0

**Cause:** `slide_line()` compared each tile with `result[-1]` even when that tile had just been produced by a merge.

**Fix:** a `just_merged` flag stops a merged tile from merging again in the same move. The function now also returns the points and number of merges, which the move functions add to the score.

```
@staticmethod
def slide_line(line):
    """Slide a line towards index 0. Returns (new_line, points, merges)."""
    values = [x for x in line if x]
    result = []
    points = 0
    merges = 0
    just_merged = False # was result[-1] created by a merge in this move?
    for value in values:
        if result and result[-1] == value and not just_merged:
            result[-1] *= 2
            points += result[-1]
            merges += 1
            just_merged = True
        else:
            result.append(value)
            just_merged = False
    return result + [0] * (SIZE - len(result)), points, merges
```

**Result:** the same row now gives the standard 2048 result 4 4 8, and the score increases by 4. Four equal tiles 2 2 2 2 give 4 4, and separated pairs such as 2 \_ 2 \_ give 4, in all four directions.

Updated code - start

```
+-----+-----+-----+-----+
| 2 | 2 | 4 | 8 |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 0 Best: 0
```

Updated code -> a (left)

```
+-----+-----+-----+-----+
| 4 | 4 | 8 |   |
+-----+-----+-----+
|   |   |   | 2 |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 4 Best: 4
Moved left: 1 merge, +4 points.
```

### 3. Task 2 - Game-State Handling

Added `Board.has_won()` (any tile of 2048 or more) and used it in the game loop together with `can_move()`. The game ends when 2048 is reached or when a full board has no equal neighbours. `Game.move()` only adds a new tile when the move actually changed the board.

Win: start

```
+-----+-----+-----+-----+
| 1024 | 1024 |   |   |
+-----+-----+-----+
| 8 | 4 |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 0 Best: 0
```

Win: > a (left)

```
+-----+-----+-----+-----+
| 2048 |   |   |   |
+-----+-----+-----+
| 8 | 4 |   | 2 |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 2048 Best: 2048
Moved left: 1 merge, +2048 points.
You reached 2048! You win.
```

No legal moves: start

```
+-----+-----+-----+-----+
| 2 | 4 | 2 | 4 |
+-----+-----+-----+
| 4 | 2 | 4 | 2 |
+-----+-----+-----+
| 2 | 4 | 2 | 4 |
+-----+-----+-----+
| 4 | 2 | 4 | 2 |
+-----+-----+-----+
Score: 0 Best: 0
No legal moves remain. Game over.
```

Unchanged move: start

```
+-----+-----+-----+-----+
| 2 | 4 |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 0 Best: 0
```

Unchanged move: > a (left)

```
+-----+-----+-----+-----+
| 2 | 4 |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
|   |   |   |   |
+-----+-----+-----+
Score: 0 Best: 0
Can't move left - nothing changed.
```

In the unchanged-move case the board is identical before and after, no tile is added, and the score is unchanged.

## 4. Task 3 - Undo and Best Score

Before each move the grid and score are saved with `Board.snapshot()`. The snapshot is kept in history only if the move changed the board, so undo reverses exactly one successful move. U restores both the board and the score without adding a tile, and a second undo reports that there is nothing to undo. The best score is updated after each successful move and is not lowered by undo.

| > a (left)                                                                                                                                                                                             | > u (undo)                                                                                                                                                                              | > u (again)                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> +-----+   4   4   8       +-----+               2   +-----+                   +-----+                   +-----+                   +-----+ Score: 4 Best: 4 Moved left: 1 merge, +4 points.</pre> | <pre> +-----+   2   2   4   8   +-----+                   +-----+                   +-----+                   +-----+                   +-----+ Score: 0 Best: 4 Undid last move.</pre> | <pre> +-----+   2   2   4   8   +-----+                   +-----+                   +-----+                   +-----+                   +-----+ Score: 0 Best: 4 Nothing to undo.</pre> |

```

def move(self, key):
    ...
    before = self.board.snapshot()
    changed, points, merges = moves[key]()
    if not changed:
        self.message = f"Can't move {DIRECTIONS[key]} - nothing changed."
        return False
    self.history = [before]          # one-level undo
    self.board.add_random_tile()
    self.best_score = max(self.best_score, self.board.score)
    ...

def undo(self):
    if not self.history:
        self.message = "Nothing to undo."
        return False
    self.board.restore(self.history.pop())
    self.message = "Undid last move."
    return True
```

## 5. Task 4 - Move Feedback

Each accepted command produces exactly one message, shown under the score. Messages are set only in `Game.move()` and `Game.undo()`, so `can_move()` and the board-scanning code never trigger feedback.

| Situation                 | Message                                    |
|---------------------------|--------------------------------------------|
| Move with merges          | Moved left: 1 merge, +4 points.            |
| Move without merges       | Moved left.                                |
| Move that changes nothing | Can't move left - nothing changed.         |
| Undo                      | Undid last move. / Nothing to undo.        |
| Invalid command           | Use W/A/S/D to move, U to undo, Q to quit. |
| Quit                      | Quit. Final score: ... Best: ...           |

```

+-----+-----+-----+-----+
Score: 4 Best: 4
Moved left.
> s

+-----+-----+-----+-----+
|         |         |         |         |
+-----+-----+-----+-----+
|         |         |         | 4       |
+-----+-----+-----+-----+
| 2       |         |         |         |
+-----+-----+-----+-----+
| 4       | 2       | 2       |         |
+-----+-----+-----+-----+
Score: 4 Best: 4
Moved down.
> a

+-----+-----+-----+-----+
|         |         |         |         |
+-----+-----+-----+-----+
| 4       |         |         |         |
+-----+-----+-----+-----+
| 2       |         | 2       |         |
+-----+-----+-----+-----+
| 4       | 4       |         |         |
+-----+-----+-----+-----+
Score: 8 Best: 8
Moved left: 1 merge, +4 points.
>

```

### Screenshot from the After video.

Each accepted move prints one action-level message. A move without merges prints Moved down., and a move with a merge prints Moved left: 1 merge, +4 points. Score and Best both increase.

## 6. Other Bugs Found and Fixed

| Bug                   | Original behaviour                                       | Fix                                                |
|-----------------------|----------------------------------------------------------|----------------------------------------------------|
| Score never increased | Score stayed at 0 after every merge                      | Move functions add merge points to the score       |
| Substring input check | key not in "wasd" accepted wa and an empty line silently | Input is checked against the exact keys W, A, S, D |
| Undo missing          | Printed Undo is not implemented yet.                     | One-level undo (Task 3)                            |

Original code - > wa (accepted, no error)

Updated code - > wa

|                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> +-----+-----+-----+-----+   16                                    +-----+-----+-----+-----+             2                             +-----+-----+-----+-----+                                           +-----+-----+-----+-----+                                           +-----+-----+-----+-----+ Score: 0 Best: 0 </pre> | <pre> +-----+-----+-----+-----+   2       4                           +-----+-----+-----+-----+                                           +-----+-----+-----+-----+                                           +-----+-----+-----+-----+                                           +-----+-----+-----+-----+ Score: 0 Best: 0 Use W/A/S/D to move, U to undo, Q to quit. </pre> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7. Testing

All cases from the lab's required-testing list were checked with an automated script against the updated code (all passed) and against the original code (fails on the merge test).

| Test case            | Input                             | Expected result            | Result |
|----------------------|-----------------------------------|----------------------------|--------|
| Ordinary slide       | 0 0 0 2, left                     | 2 0 0 0                    | Pass   |
| Four equal tiles     | 2 2 2 2, left / right / up / down | 4 4 (once each), +8 points | Pass   |
| Repeated pattern     | 2 2 4 8, left                     | 4 4 8 (no chain merge)     | Pass   |
| Separated equal pair | 2 0 2 0, left                     | 4 0 0 0                    | Pass   |
| No merge possible    | 2 4 2 4, left                     | Unchanged                  | Pass   |

| Test case        | Input                    | Expected result                                   | Result |
|------------------|--------------------------|---------------------------------------------------|--------|
| Unchanged move   | 2 4 __, left             | Board, score and history unchanged; no tile added | Pass   |
| 2048 detection   | 1024 1024, left          | Game ends with win message                        | Pass   |
| No-move board    | Full checkerboard of 2/4 | Game ends with no-moves message                   | Pass   |
| Undo             | Move, then U             | Board and score restored, no tile added           | Pass   |
| Undo twice       | Move, U, U               | Second undo: Nothing to undo                      | Pass   |
| Best score       | Move (+4), then U        | Score back to 0, Best stays 4                     | Pass   |
| Invalid commands | x, wa, empty line        | Error message each time, no move                  | Pass   |
| Quit             | Q                        | Game exits with final score                       | Pass   |

## 8. Prompts Used

The complete chat history is at <https://claude.ai/share/463f54ba-b87c-44e4-aa29-bf588b3ee43b>. One prompt was written per task, each including only the relevant block of code:

| # | Task             | Prompt (summary)                                                                                                                                                               |
|---|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Merge bug        | Pasted slide_line() and move_left(). Asked to stop a just-merged tile merging again (2 2 4 8 should give 4 4 8), and to return points and merge counts so the score increases. |
| 2 | Game state       | Pasted Game.move() and the win/no-move checks in run(). Asked for Board.has_won(), ending on 2048 or no moves, and no new tile after an unchanged move.                        |
| 3 | Undo, best score | Pasted the U handler in run(). Asked for a one-level undo restoring board and score without adding a tile, and best score tracking.                                            |
| 4 | Move feedback    | Pasted display() and the input check. Asked for one feedback message per command, set only in move()/undo(), and an exact-key input check instead of the substring check.      |